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Education

National Institute of Technology Karnataka, Surathkal

2023 – Present

B.Tech in Civil Engineering, CGPA: 7.92

Minor in Electronics & Communication, CGPA: 7.25

Experience

Web Lead - Osdag, IIT Bombay *React, Django, Redis, WebSockets*

May 2025 – Present

- Led a 4-member web team on production modules (Tension Members, Base Plate Connection) and a frontend/backend refactor adopted as the team standard; my UI redesign was chosen over other interns' designs.
- Architected **Redis caching** and async CAD computation with Celery, plus **WebSocket** infrastructure for real-time job status and collaboration; owned bare-metal deployment at IIT Bombay (Nginx, SSL, CI/CD).
- Wrote load-testing scripts and a live CPU/RAM/core dashboard, coordinating 20+ members to stress-test **200+ concurrent requests**; used the results to document the recommended production server configuration.

App Dev - IRIS, NITK *Dart, Flutter, BLoC*

Nov 2024 – Present

- Extended IRIS, NITK's student-built ERP with **55+ digitized** processes, by shipping 4 missing modules: **Event Management, Alumni Donations**, live **Hostel League** tracking and **Service Requests**.
- Architected Event Management with Clean Architecture and BLoC on Flutter + NoSQL for layered, testable code.
- Integrated the BillDesk payment SDK for alumni donations and built real-time hostel league tracking for students.

App Dev - Humora Technologies *Flutter, Cloud Functions, Riverpod*

Nov 2025 – Feb 2026

- Built a paginated event feed with real-time status filters and per-author notification subscriptions.
- Implemented multi-level role-based access with Firestore Security Rules, enforcing admin promotion server-side.
- Added photo and video likes with a creator filter to sort media by who liked it, using Riverpod for state.

Key Projects

Process Mining & Analytics Platform

Java, Spring Boot, PostgreSQL, JPA, Hibernate

- Built a Spring Boot backend ingesting CSV event logs into PostgreSQL (2,500+ cases, 16K+ events), using hash-based grouping, sorting and graph construction to discover Directly-Follows Graphs and variants.
- Exposed 8+ REST analytics APIs for P50/P90/P95 latency, throughput, bottlenecks and rework, surfacing planted patterns like a ~7.7-hour payment-to-packing bottleneck and ~22% rework rate.
- Implemented conformance checking via dynamic-programming edit-distance alignment with traceback, flagging skipped, extra, repeated and out-of-order activities and computing case-level fitness.

OpenLake - Distributed Data Platform

Spark, Kafka, Delta Lake, FastAPI, Docker, Terraform, Azure

- Designed a distributed processing platform on Spark and Delta Lake over S3-compatible storage, orchestrated by Airflow, with idempotent MERGE upserts and an automated validation gate for bad input.
- Deployed to Azure with Terraform (ADLS Gen2, Azure SQL); when a Databricks vCPU quota blocked compute, re-architected to decouple compute from storage (**s3a** → **abfss**), keeping one codebase portable.
- Built an event-driven path with Kafka (Redpanda) and Spark Structured Streaming for live order metrics, and served ML predictions via a FastAPI service with batched database writes.

SysSight - Distributed System Monitoring System

Python, FastAPI, TimescaleDB, Docker

- Built a distributed monitoring system where lightweight Python agents on each host collect CPU, memory and disk metrics at high frequency and report them to a central FastAPI server.
- Stored metrics in TimescaleDB (a PostgreSQL time-series extension) for fast time-range queries across multiple hosts, with the full stack containerized in Docker for portable deployment.
- Designed event-driven alerting on user-configurable thresholds with server-side deduplication, so a sustained spike sends one Discord notification instead of flooding the channel.

Technical Skills

Languages	Java, Python, C++, JavaScript, Dart, SQL
Frameworks	Spring Boot, Hibernate/JPA, FastAPI, Django, React, Flutter (BLoC, Riverpod)
Data & Infra	PostgreSQL, TimescaleDB, MongoDB, Redis, Kafka, Apache Spark, Delta Lake, Airflow
Cloud & Tools	Docker, Kubernetes, Terraform, Azure, Nginx, Git, GitHub Actions, Linux

Achievements and Awards

- Shortlisted for the upcoming final round of EthGlobal New Delhi, an **international level Hackathon**.
- Secured **3rd place out of 190+** participants in the AssetHub Hackathon (Goa, 2025) and was a finalist in all 4 hackathons participated.
- Secured **3rd place** in the Builder house Blr where I built a bridge between INR and Polkadot tokens